1.Writing a Simple Linux Driver and Recompiling the Kernel (50%)

(1) Setting up the Environment: Verify if kernel headers and development tools are installed on the system. (Command: uname -r and sudo apt install build-essential linux-headers-$(uname -r)). Provide your screenshot. (5 pts)

uname -r:

```
waynelin@waynelin-virtual-machine:~/Desktop$ uname -r
6.8.0-87-generic
```

sudo apt install build-essential linux-headers-$(uname -r)):

```
waynelin@waynelin-virtual-machine:~/Desktop$ sudo apt install -y build-essential linux-headers-$(uname -r)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
linux-headers-6.8.0-87-generic is already the newest version (6.8.0-87.88~22.04.1).
linux-headers-6.8.0-87-generic set to manually installed.
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu dpkg-dev fakeroot g++
  g++-11 gcc gcc-11 libalgorithm-diff-perl libalgorithm-diff-xs-perl
  libalgorithm-merge-perl libasan6 libbinutils libc-dev-bin libc-devtools
  libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0 libctf0 libdpkg-perl
  libfakeroot libfile-fcntllock-perl libgcc-11-dev libitm1 liblsan0 libnsl-dev
  libquadmath0 libstdc++-11-dev libtirpc-dev libtsan0 libubsan1 linux-libc-dev
  lto-disabled-list make manpages-dev rpcsvc-proto
Suggested packages:
  binutils-doc debian-keyring g++-multilib g++-11-multilib gcc-11-doc
  gcc-multilib autoconf automake libtool flex bison gcc-doc gcc-11-multilib
  gcc-11-locales glibc-doc git bzr libstdc++-11-doc make-doc
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential dpkg-dev
  fakeroot g++ g++-11 gcc gcc-11 libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libbinutils
  libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0
  libctf0 libdpkg-perl libfakeroot libfile-fcntllock-perl libgcc-11-dev
  libitm1 liblsan0 libnsl-dev libquadmath0 libstdc++-11-dev libtirpc-dev
  libtsan0 libubsan1 linux-libc-dev lto-disabled-list make manpages-dev
  rpcsvc-proto
0 upgraded, 39 newly installed, 0 to remove and 0 not upgraded.
Need to get 54.0 MB/54.2 MB of archives.
After this operation, 186 MB of additional disk space will be used.
Get:1 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc-dev-bin amd64 2.35-0ubuntu3.11 [20.3 kB]
Get:2 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-libc-dev amd64 5.15.0-161.171 [1307 kB]
Get:3 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 libcrypt-dev amd64 1:4.4.27-1 [112 kB]
Get:4 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 rpcsvc-proto amd64 1.4.2-0ubuntu6 [68.5 kB]
Get:5 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtirpc-dev amd64 1.3.2-2ubuntu0.1 [192 kB]
Get:6 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 libnsl-dev amd64 1.3.0-2build2 [71.3 kB]
Get:7 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6-dev amd64 2.35-0ubuntu3.11 [2100 kB]
Get:8 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcc1-0 amd64 12.3.0-1ubuntu1~22.04.2 [48.3 kB]
Get:9 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libitm1 amd64 12.3.0-1ubuntu1~22.04.2 [30.2 kB]
Get:10 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libasan6 amd64 11.4.0-1ubuntu1~22.04.2 [2283 kB]
```

```
Get:11 http://security.ubuntu.com/ubuntu jammy-security/main amd64 binutils-common amd64 2.38-4ubuntu2.11 [223 kB]
Get:12 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 liblsan0 amd64 12.3.0-1ubuntu1~22.04.2 [1069 kB]
Get:13 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtsan0 amd64 11.4.0-1ubuntu1~22.04.2 [2262 kB]
Get:14 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libubsan1 amd64 12.3.0-1ubuntu1~22.04.2 [976 kB]
Get:15 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libquadmath0 amd64 12.3.0-1ubuntu1~22.04.2 [154 kB]
Get:16 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgcc-11-dev amd64 11.4.0-1ubuntu1~22.04.2 [2517 kB]
Get:17 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gcc-11 amd64 11.4.0-1ubuntu1~22.04.2 [20.1 MB]
Get:18 http://security.ubuntu.com/ubuntu jammy-security/main amd64 libbinutils amd64 2.38-4ubuntu2.11 [658 kB]
Get:19 http://security.ubuntu.com/ubuntu jammy-security/main amd64 libctf-nobfd0 amd64 2.38-4ubuntu2.11 [108 kB]
Get:20 http://security.ubuntu.com/ubuntu jammy-security/main amd64 libctf0 amd64 2.38-4ubuntu2.11 [103 kB]
Get:21 http://security.ubuntu.com/ubuntu jammy-security/main amd64 binutils-x86-64-linux-gnu amd64 2.38-4ubuntu2.11 [2324 kB]
Get:22 http://security.ubuntu.com/ubuntu jammy-security/main amd64 binutils amd64 2.38-4ubuntu2.11 [3200 B]
Get:23 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 gcc amd64 4:11.2.0-1ubuntu1 [5112 B]
Get:24 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libstdc++-11-dev amd64 11.4.0-1ubuntu1~22.04.2 [2101 kB]
Get:25 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 g++-11 amd64 11.4.0-1ubuntu1~22.04.2 [11.4 MB]
Get:26 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 g++ amd64 4:11.2.0-1ubuntu1 [1412 B]
Get:27 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 make amd64 4.3-4.1build1 [180 kB]
Get:28 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 lto-disabled-list all 24 [12.5 kB]
Get:29 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 dpkg-dev all 1.21.1ubuntu2.6 [922 kB]
Get:30 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 build-essential amd64 12.9ubuntu3 [4744 B]
Get:31 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 libfakeroot amd64 1.28-1ubuntu1 [31.5 kB]
Get:32 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 fakeroot amd64 1.28-1ubuntu1 [60.4 kB]
Get:33 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 libalgorithm-diff-perl all 1.201-1 [41.8 kB]
Get:34 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 libalgorithm-diff-xs-perl amd64 0.04-6build3 [11.9 kB]
Get:35 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 libalgorithm-merge-perl all 0.08-3 [12.0 kB]
Get:36 http://tw.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc-devtools amd64 2.35-0ubuntu3.11 [29.0 kB]
Get:37 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 libfile-fcntllock-perl amd64 0.22-3build7 [33.9 kB]
Get:38 http://tw.archive.ubuntu.com/ubuntu jammy/main amd64 manpages-dev all 5.10-1ubuntu1 [2309 kB]
Fetched 54.0 MB in 6s (9179 kB/s)
Extracting templates from packages: 100%
Selecting previously unselected package binutils-common:amd64.
(Reading database ... 202582 files and directories currently installed.)
Preparing to unpack .../00-binutils-common_2.38-4ubuntu2.11_amd64.deb ...
Unpacking binutils-common:amd64 (2.38-4ubuntu2.11) ...
Selecting previously unselected package libbinutils:amd64.
Preparing to unpack .../01-libbinutils_2.38-4ubuntu2.11_amd64.deb ...
Unpacking libbinutils:amd64 (2.38-4ubuntu2.11) ...
Selecting previously unselected package libctf-nobfd0:amd64.
Preparing to unpack .../02-libctf-nobfd0_2.38-4ubuntu2.11_amd64.deb ...
Unpacking libctf-nobfd0:amd64 (2.38-4ubuntu2.11) ...
Selecting previously unselected package libctf0:amd64.
Preparing to unpack .../03-libctf0_2.38-4ubuntu2.11_amd64.deb ...
Unpacking libctf0:amd64 (2.38-4ubuntu2.11) ...
Selecting previously unselected package binutils-x86-64-linux-gnu.
Preparing to unpack .../04-binutils-x86-64-linux-gnu_2.38-4ubuntu2.11_amd64.deb
```

```
...
Unpacking binutils-x86-64-linux-gnu (2.38-4ubuntu2.11) ...
Selecting previously unselected package binutils.
Preparing to unpack .../05-binutils_2.38-4ubuntu2.11_amd64.deb ...
Unpacking binutils (2.38-4ubuntu2.11) ...
Selecting previously unselected package libc-dev-bin.
Preparing to unpack .../06-libc-dev-bin_2.35-0ubuntu3.11_amd64.deb ...
Unpacking libc-dev-bin (2.35-0ubuntu3.11) ...
Selecting previously unselected package linux-libc-dev:amd64.
Preparing to unpack .../07-linux-libc-dev_5.15.0-161.171_amd64.deb ...
Unpacking linux-libc-dev:amd64 (5.15.0-161.171) ...
Selecting previously unselected package libcrypt-dev:amd64.
Preparing to unpack .../08-libcrypt-dev_1%3a4.4.27-1_amd64.deb ...
Unpacking libcrypt-dev:amd64 (1:4.4.27-1) ...
Selecting previously unselected package rpcsvc-proto.
Preparing to unpack .../09-rpcsvc-proto_1.4.2-0ubuntu6_amd64.deb ...
Unpacking rpcsvc-proto (1.4.2-0ubuntu6) ...
Selecting previously unselected package libtirpc-dev:amd64.
Preparing to unpack .../10-libtirpc-dev_1.3.2-2ubuntu0.1_amd64.deb ...
Unpacking libtirpc-dev:amd64 (1.3.2-2ubuntu0.1) ...
Selecting previously unselected package libnsl-dev:amd64.
Preparing to unpack .../11-libnsl-dev_1.3.0-2build2_amd64.deb ...
Unpacking libnsl-dev:amd64 (1.3.0-2build2) ...
Selecting previously unselected package libc6-dev:amd64.
Preparing to unpack .../12-libc6-dev_2.35-0ubuntu3.11_amd64.deb ...
Unpacking libc6-dev:amd64 (2.35-0ubuntu3.11) ...
Selecting previously unselected package libcc1-0:amd64.
Preparing to unpack .../13-libcc1-0_12.3.0-1ubuntu1~22.04.2_amd64.deb ...
Unpacking libcc1-0:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package libitm1:amd64.
Preparing to unpack .../14-libitm1_12.3.0-1ubuntu1~22.04.2_amd64.deb ...
Unpacking libitm1:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package libasan6:amd64.
Preparing to unpack .../15-libasan6_11.4.0-1ubuntu1~22.04.2_amd64.deb ...
Unpacking libasan6:amd64 (11.4.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package liblsan0:amd64.
Preparing to unpack .../16-liblsan0_12.3.0-1ubuntu1~22.04.2_amd64.deb ...
Unpacking liblsan0:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package libtsan0:amd64.
Preparing to unpack .../17-libtsan0_11.4.0-1ubuntu1~22.04.2_amd64.deb ...
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```
Selecting previously unselected package libubsan1:amd64.
Preparing to unpack .../18-libubsan1_12.3.0-1ubuntu1~22.04.2_amd64.deb ...
Unpacking libubsan1:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package libquadmath0:amd64.
Preparing to unpack .../19-libquadmath0_12.3.0-1ubuntu1~22.04.2_amd64.deb ...
Unpacking libquadmath0:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package libgcc-11-dev:amd64.
Preparing to unpack .../20-libgcc-11-dev_11.4.0-1ubuntu1~22.04.2_amd64.deb ...
Unpacking libgcc-11-dev:amd64 (11.4.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package gcc-11.
Preparing to unpack .../21-gcc-11_11.4.0-1ubuntu1~22.04.2_amd64.deb ...
Unpacking gcc-11 (11.4.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package gcc.
Preparing to unpack .../22-gcc_4%3a11.2.0-1ubuntu1_amd64.deb ...
Unpacking gcc (4:11.2.0-1ubuntu1) ...
Selecting previously unselected package libstdc++-11-dev:amd64.
Preparing to unpack .../23-libstdc++-11-dev_11.4.0-1ubuntu1~22.04.2_amd64.deb ..
.
Unpacking libstdc++-11-dev:amd64 (11.4.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package g++-11.
Preparing to unpack .../24-g++-11_11.4.0-1ubuntu1~22.04.2_amd64.deb ...
Unpacking g++-11 (11.4.0-1ubuntu1~22.04.2) ...
Selecting previously unselected package g++.
Preparing to unpack .../25-g++_4%3a11.2.0-1ubuntu1_amd64.deb ...
Unpacking g++ (4:11.2.0-1ubuntu1) ...
Selecting previously unselected package make.
Preparing to unpack .../26-make_4.3-4.1build1_amd64.deb ...
Unpacking make (4.3-4.1build1) ...
Selecting previously unselected package libdpkg-perl.
Preparing to unpack .../27-libdpkg-perl_1.21.1ubuntu2.6_all.deb ...
Unpacking libdpkg-perl (1.21.1ubuntu2.6) ...
Selecting previously unselected package lto-disabled-list.
Preparing to unpack .../28-lto-disabled-list_24_all.deb ...
Unpacking lto-disabled-list (24) ...
Selecting previously unselected package dpkg-dev.
Preparing to unpack .../29-dpkg-dev_1.21.1ubuntu2.6_all.deb ...
Unpacking dpkg-dev (1.21.1ubuntu2.6) ...
Selecting previously unselected package build-essential.
Preparing to unpack .../30-build-essential_12.9ubuntu3_amd64.deb ...
Unpacking build-essential (12.9ubuntu3) ...
Selecting previously unselected package libfakeroot:amd64.
Preparing to unpack .../31-libfakeroot_1.28-1ubuntu1_amd64.deb ...
Unpacking libfakeroot:amd64 (1.28-1ubuntu1) ...
Selecting previously unselected package fakeroot.
```

```
Preparing to unpack .../32-fakeroot_1.28-1ubuntu1_amd64.deb ...
Unpacking fakeroot (1.28-1ubuntu1) ...
Selecting previously unselected package libalgorithm-diff-perl.
Preparing to unpack .../33-libalgorithm-diff-perl_1.201-1_all.deb ...
Unpacking libalgorithm-diff-perl (1.201-1) ...
Selecting previously unselected package libalgorithm-diff-xs-perl.
Preparing to unpack .../34-libalgorithm-diff-xs-perl_0.04-6build3_amd64.deb ...
Unpacking libalgorithm-diff-xs-perl (0.04-6build3) ...
Selecting previously unselected package libalgorithm-merge-perl.
Preparing to unpack .../35-libalgorithm-merge-perl_0.08-3_all.deb ...
Unpacking libalgorithm-merge-perl (0.08-3) ...
Selecting previously unselected package libc-devtools.
Preparing to unpack .../36-libc-devtools_2.35-0ubuntu3.11_amd64.deb ...
Unpacking libc-devtools (2.35-0ubuntu3.11) ...
Selecting previously unselected package libfile-fcntllock-perl.
Preparing to unpack .../37-libfile-fcntllock-perl_0.22-3build7_amd64.deb ...
Unpacking libfile-fcntllock-perl (0.22-3build7) ...
Selecting previously unselected package manpages-dev.
Preparing to unpack .../38-manpages-dev_5.10-1ubuntu1_all.deb ...
Unpacking manpages-dev (5.10-1ubuntu1) ...
Setting up manpages-dev (5.10-1ubuntu1) ...
Setting up lto-disabled-list (24) ...
Setting up libfile-fcntllock-perl (0.22-3build7) ...
Setting up libalgorithm-diff-perl (1.201-1) ...
Setting up binutils-common:amd64 (2.38-4ubuntu2.11) ...
Setting up linux-libc-dev:amd64 (5.15.0-161.171) ...
Setting up libctf-nobfd0:amd64 (2.38-4ubuntu2.11) ...
Setting up libfakeroot:amd64 (1.28-1ubuntu1) ...
Setting up libasan6:amd64 (11.4.0-1ubuntu1~22.04.2) ...
Setting up fakeroot (1.28-1ubuntu1) ...
update-alternatives: using /usr/bin/fakeroot-sysv to provide /usr/bin/fakeroot (
fakeroot) in auto mode
```

```
Setting up libtirpc-dev:amd64 (1.3.2-2ubuntu0.1) ...
Setting up rpcsvc-proto (1.4.2-0ubuntu6) ...
Setting up make (4.3-4.1build1) ...
Setting up libquadmath0:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Setting up libdpkg-perl (1.21.1ubuntu2.6) ...
Setting up libubsan1:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Setting up libnsl-dev:amd64 (1.3.0-2build2) ...
Setting up libcrypt-dev:amd64 (1:4.4.27-1) ...
Setting up libbinutils:amd64 (2.38-4ubuntu2.11) ...
Setting up libc-dev-bin (2.35-0ubuntu3.11) ...
Setting up libalgorithm-diff-xs-perl (0.04-6build3) ...
Setting up libcc1-0:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Setting up liblsan0:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Setting up libitm1:amd64 (12.3.0-1ubuntu1~22.04.2) ...
Setting up libc-devtools (2.35-0ubuntu3.11) ...
Setting up libalgorithm-merge-perl (0.08-3) ...
Setting up libtsan0:amd64 (11.4.0-1ubuntu1~22.04.2) ...
Setting up libctf0:amd64 (2.38-4ubuntu2.11) ...
Setting up libgcc-11-dev:amd64 (11.4.0-1ubuntu1~22.04.2) ...
Setting up libc6-dev:amd64 (2.35-0ubuntu3.11) ...
Setting up binutils-x86-64-linux-gnu (2.38-4ubuntu2.11) ...
Setting up binutils (2.38-4ubuntu2.11) ...
Setting up dpkg-dev (1.21.1ubuntu2.6) ...
Setting up libstdc++-11-dev:amd64 (11.4.0-1ubuntu1~22.04.2) ...
Setting up gcc-11 (11.4.0-1ubuntu1~22.04.2) ...
Setting up g++-11 (11.4.0-1ubuntu1~22.04.2) ...
Setting up gcc (4:11.2.0-1ubuntu1) ...
Setting up g++ (4:11.2.0-1ubuntu1) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mo
de
Setting up build-essential (12.9ubuntu3) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...
```

(2) Prepare a simple “Hello World” kernel module. You are asked to write a simple “Hello World” kernel module and a Makefile to compile your kernel module. (10 pts)

hello.c:

```c
// hello.c
#include <linux/init.h>
#include <linux/module.h>
#include <linux/kernel.h>

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Wayne");     // 把這裡改成你的名字
MODULE_DESCRIPTION("A simple Hello World module");

static int __init hello_init(void)
{
    pr_info("hello_kmod: Hello, world!\n");
    return 0;
}

static void __exit hello_exit(void)
{
    pr_info("hello_kmod: Goodbye, world!\n");
}

module_init(hello_init);
module_exit(hello_exit);
```

Makefile:

```makefile
obj-m += hello.o

KDIR := /lib/modules/$(shell uname -r)/build
PWD  := $(shell pwd)

all:
        $(MAKE) -C $(KDIR) M=$(PWD) modules

clean:
        $(MAKE) -C $(KDIR) M=$(PWD) clean
```

Make 結果:

```
waynelin@waynelin-virtual-machine:~/kmod_hello$ make
make -C /lib/modules/6.8.0-87-generic/build M=/home/waynelin/kmod_hello modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-87-generic'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.2) 12.3.0
  You are using:           gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.2) 12.3.0
  CC [M]  /home/waynelin/kmod_hello/hello.o
  MODPOST /home/waynelin/kmod_hello/Module.symvers
  CC [M]  /home/waynelin/kmod_hello/hello.mod.o
  LD [M]  /home/waynelin/kmod_hello/hello.ko
  BTF [M] /home/waynelin/kmod_hello/hello.ko
Skipping BTF generation for /home/waynelin/kmod_hello/hello.ko due to unavailability of vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-87-generic'
waynelin@waynelin-virtual-machine:~/kmod_hello$ ls
hello.c  hello.ko  hello.mod  hello.mod.c  hello.mod.o  hello.o  Makefile  modules.order  Module.symvers
```

(3) Loading and unloading the module. After compiling your module, you should use commands (i.e., insmod and rmmod) to load and unload your module. Moreover, you need to verify the module status by checking the kernel log (command: dmesg). (10 pts)

Insmod 後:

```
waynelin@waynelin-virtual-machine:~/kmod_hello$ sudo insmod hello.ko
[sudo] password for waynelin:
waynelin@waynelin-virtual-machine:~/kmod_hello$ lsmod | grep hello
hello                  12288  0
```

```
waynelin@waynelin-virtual-machine:~/kmod_hello$ sudo dmesg | tail -n 20
[42985.367931] audit: type=1400 audit(1764652825.129:68): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.firefox.firefox" pid=5628 comm="
apparmor_parser"
[42985.438397] audit: type=1400 audit(1764652825.199:69): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.firefox.geckodriver" pid=5629 co
mm="apparmor_parser"
[42985.537431] audit: type=1400 audit(1764652825.297:70): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.hook.configure" pid=5
634 comm="apparmor_parser"
[42986.001136] audit: type=1400 audit(1764652825.760:71): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap-update-ns.snap-store" pid=5633 c
omm="apparmor_parser"
[42986.213316] audit: type=1400 audit(1764652825.975:72): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=5635
comm="apparmor_parser"
[42986.281099] audit: type=1400 audit(1764652826.041:73): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=
5636 comm="apparmor_parser"
[43010.421948] kauditd_printk_skb: 4 callbacks suppressed
[43010.421952] audit: type=1400 audit(1764652850.182:78): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" na
me="/snap/snapd/21759/usr/lib/snapd/snap-confine" pid=5655 comm="apparmor_parser"
[43010.421958] audit: type=1400 audit(1764652850.182:79): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" na
me="/snap/snapd/21759/usr/lib/snapd/snap-confine//mount-namespace-capture-helper" pid=5655 comm="apparmor_parser"
[43011.229599] audit: type=1400 audit(1764652850.990:80): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snapd-desktop-integration.snapd-
desktop-integration" pid=5658 comm="apparmor_parser"
[43011.589246] audit: type=1400 audit(1764652851.350:81): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" na
me="snap-update-ns.snapd-desktop-integration" pid=5657 comm="apparmor_parser"
[43011.624011] audit: type=1400 audit(1764652851.384:82): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" na
me="snap.snapd-desktop-integration.hook.configure" pid=5660 comm="apparmor_parser"
[43015.081532] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 64 times, consider switching to WQ_UNBOUND
[51313.322052] workqueue: mld_ifc_work hogged CPU for >10000us 8 times, consider switching to WQ_UNBOUND
[63934.116906] INFO: NMI handler (perf_event_nmi_handler) took too long to run: 21.415 msecs
[63934.125663] perf: interrupt took too long (4826 > 2500), lowering kernel.perf_event_max_sample_rate to 41000
[66008.728366] workqueue: psi_avgs_work hogged CPU for >10000us 16 times, consider switching to WQ_UNBOUND
[66231.091042] hello: loading out-of-tree module taints kernel.
[66231.092855] hello: module verification failed: signature and/or required key missing - tainting kernel
[66231.286545] hello_kmod: Hello, world!
```

rmmod 後:

```
waynelin@waynelin-virtual-machine:~/kmod_hello$ sudo rmmod hello
waynelin@waynelin-virtual-machine:~/kmod_hello$ lsmod | grep hello
waynelin@waynelin-virtual-machine:~/kmod_hello$ sudo dmesg | tail -n 20
[66932.661940] audit: type=1400 audit(1764676772.937:89): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" na
me="snap.firefox.hook.configure" pid=6293 comm="apparmor_parser"
[66932.667914] audit: type=1400 audit(1764676772.943:90): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" na
me="snap.firefox.hook.post-refresh" pid=6295 comm="apparmor_parser"
[66932.679495] audit: type=1400 audit(1764676772.953:91): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" na
me="snap.snap-store.hook.configure" pid=6296 comm="apparmor_parser"
[66932.822199] audit: type=1400 audit(1764676773.097:92): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" na
me="snap.firefox.firefox" pid=6291 comm="apparmor_parser"
[66937.139476] loop11: detected capacity change from 0 to 8
[66938.390660] kauditd_printk_skb: 6 callbacks suppressed
[66938.390665] audit: type=1400 audit(1764676778.659:99): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" na
me="/usr/lib/snapd/snap-confine" pid=6357 comm="apparmor_parser"
[66938.424856] audit: type=1400 audit(1764676778.700:100): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" n
ame="/usr/lib/snapd/snap-confine//mount-namespace-capture-helper" pid=6357 comm="apparmor_parser"
[66965.527948] loop11: detected capacity change from 0 to 513184
[66967.628776] audit: type=1400 audit(1764676807.903:101): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="/snap/snapd/25577/usr/lib/snapd/snap
-confine" pid=6492 comm="apparmor_parser"
[66967.635502] audit: type=1400 audit(1764676807.911:102): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="/snap/snapd/25577/usr/lib/snapd/snap
-confine//mount-namespace-capture-helper" pid=6492 comm="apparmor_parser"
[66968.000056] audit: type=1400 audit(1764676808.274:103): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.firefox.hook.configure" pid=649
7 comm="apparmor_parser"
[66968.218337] audit: type=1400 audit(1764676808.493:104): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.firefox.hook.disconnect-plug-ho
st-hunspell" pid=6498 comm="apparmor_parser"
[66968.425458] audit: type=1400 audit(1764676808.700:105): apparmor="STATUS" operation="profile_load" profile="unconfined" name="snap.firefox.hook.install" pid=6499 com
m="apparmor_parser"
[66968.670033] audit: type=1400 audit(1764676808.945:106): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.firefox.hook.post-refresh" pid=
6500 comm="apparmor_parser"
[66968.689264] audit: type=1400 audit(1764676808.965:107): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap-update-ns.firefox" pid=6494 com
m="apparmor_parser"
[66968.838645] audit: type=1400 audit(1764676809.115:108): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.firefox.firefox" pid=6495 comm=
"apparmor_parser"
[66968.892337] audit: type=1400 audit(1764676809.167:109): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.firefox.geckodriver" pid=6496 c
omm="apparmor_parser"
[66969.177248] audit: type=1400 audit(1764676809.452:110): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" n
ame="/snap/snapd/25577/usr/lib/snapd/snap-confine" pid=6523 comm="apparmor_parser"
[67103.284352] hello_kmod: Goodbye, world!
```

(4) Recompiling the kernel to include the module. Students are asked to clone the Linux kernel source and reconfigure the kernel to build the module as a built-in part of the kernel (not a loadable module). Verify the kernel includes the module by checking dmesg. (15 pts)

先透過 **make menuconfig** 修改 **Kconfig**，將 **module** 真得改成 **build-in** 的，並重新開機確認:

```
waynelin@waynelin-virtual-machine:~/Desktop$ uname -r
6.18.0-hello+
```

驗證 built-in driver 有被編進 kernel 並在開機時執行:

```
waynelin@waynelin-virtual-machine:~/Desktop$ sudo dmesg | grep -i hello
[sudo] password for waynelin:
[    0.000000] Linux version 6.18.0-hello+ (waynelin@waynelin-virtual-machine) (
gcc (Ubuntu 11.4.0-1ubuntu1~22.04.2) 11.4.0, GNU ld (GNU Binutils for Ubuntu) 2.
38) #1 SMP PREEMPT_DYNAMIC Tue Dec  2 22:06:40 CST 2025
[    0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-6.18.0-hello+ root=UUID=15
cc4143-fec2-4406-bd6f-1875053da18e ro find_preseed=/preseed.cfg auto noprompt pr
iority=critical locale=en_US quiet splash
[    0.454473] Kernel command line: BOOT_IMAGE=/boot/vmlinuz-6.18.0-hello+ root=
UUID=15cc4143-fec2-4406-bd6f-1875053da18e ro find_preseed=/preseed.cfg auto nopr
ompt priority=critical locale=en_US quiet splash
[    1.892945] hello_kmod: Hello, world!
[    1.921874] usb usb1: Manufacturer: Linux 6.18.0-hello+ ehci_hcd
[    1.923025] usb usb2: Manufacturer: Linux 6.18.0-hello+ uhci_hcd
[   27.659400] CPU: 3 UID: 0 PID: 145 Comm: kworker/u523:0 Not tainted 6.18.0-he
llo+ #1 PREEMPT(voluntary)
[   54.090972] CPU: 1 UID: 0 PID: 477 Comm: plymouthd Tainted: G        W
    6.18.0-hello+ #1 PREEMPT(voluntary)
```

lsmod | grep hello 無任何輸出，但在 dmesg 中可以看到 hello_kmod: Hello, world! 在開機初期即被印出，因此可確認此 driver 以 **built-in** 形式編入核心而非動態載入模組:

```
waynelin@waynelin-virtual-machine:~/Desktop$ lsmod | grep -i hello
waynelin@waynelin-virtual-machine:~/Desktop$
```

2. Collecting and Processing Temperature Data on MSP430 FR4133 using FRAM (50%)

(1) Setting up the temperature sensor. Students are tasked with configuring the MSP430's ADC12 to collect data from the internal temperature sensor. Implement the function readTemperature() to periodically collect temperature data. (10 pts)

initTemperatureSensor():

```
void initTemperatureSensor(void)
{
    // 先關掉 ADC，才可以改設定
    ADCCTL0 &= ~ADCENC;

    // 解鎖 PMM
    PMMCTL0_H = PMMPW_H;
    // 開啟 internal reference + temperature sensor
    PMMCTL2 |= INTREFEN | TSENSOREN;
    // 這裡用預設的參考電壓即可（對應 ADCSREF_1）
    PMMCTL0_H = 0;              // 鎖回 PMM

    // 設定 ADC
    ADCCTL0 = ADCSHT_5 | ADCON;     // 足夠長的 sample time + 開啟 ADC
    ADCCTL1 = ADCSHP;               // 使用 sampling timer
#ifdef ADCRES                         // 如果有 ADCRES，就設最高解析度
    ADCCTL2 = ADCRES;
#endif

    // 使用 A12（內部溫度感測器），參考電壓用 internal reference
    ADCMCTL0 = ADCINCH_12 | ADCSREF_1;

    // 允許 ADC 轉換
    ADCCTL0 |= ADCENC;

    // 等內部參考電壓穩定一下（1 MHz clock 約 0.1 秒）
    __delay_cycles(100000);
}
```

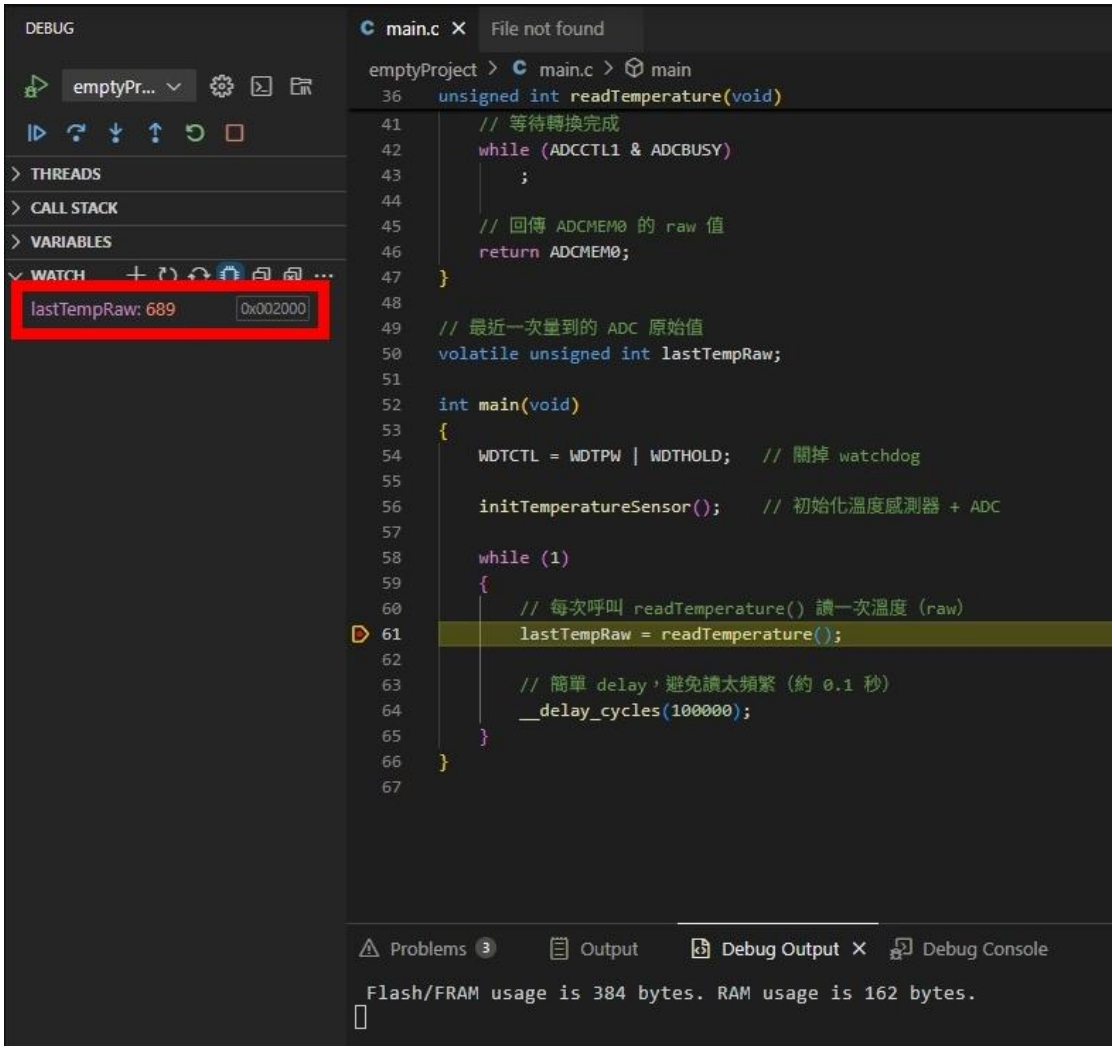
readTemperature():

```c
unsigned int readTemperature(void)
{
    // 開始轉換
    ADCCTL0 |= ADCSC;

    // 等待轉換完成
    while (ADCCTL1 & ADCBUSY)
        ;

    // 回傳 ADCMEM0 的 raw 值
    return ADCMEM0;
}
```

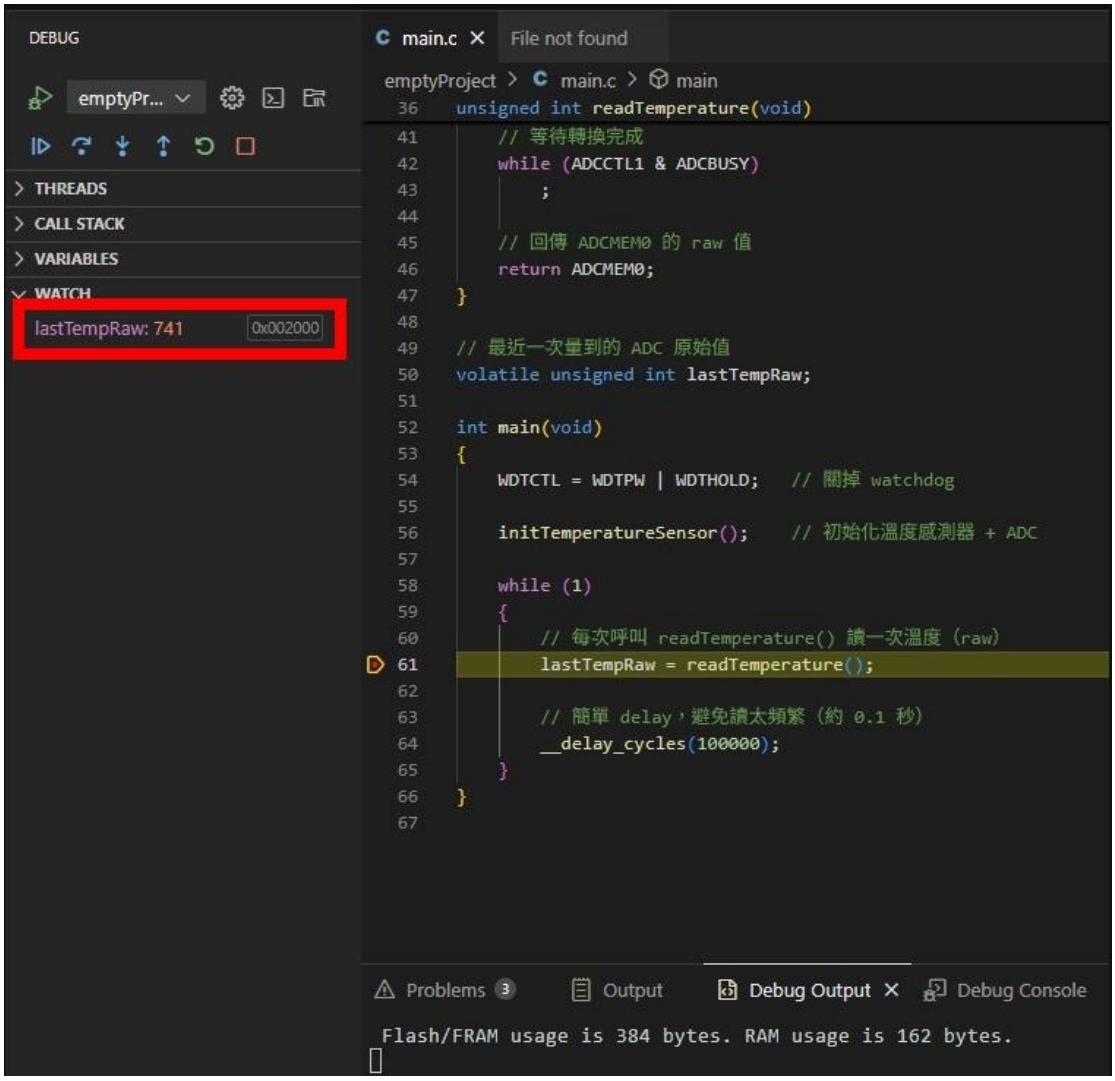
Call readTemperature() 並存到 lastTempRaw:

```c
while (1)
{
    // 每次呼叫 readTemperature() 讀一次溫度 (raw)
    lastTempRaw = readTemperature();

    // 簡單 delay，避免讀太頻繁 (約 0.1 秒)
    __delay_cycles(100000);
}
```

實作結果(平常):


DEBUG
emptyPr...
THREADS
CALL STACK
VARIABLES
WATCH
lastTempRaw: 689
0x002000
main.c
File not found
emptyProject > main.c > main
36 unsigned int readTemperature(void)
41 // 等待轉換完成
42 while (ADCCTL1 & ADCBUSY)
43 ;
44
45 // 回傳 ADCMEM0 的 raw 值
46 return ADCMEM0;
47 }
48
49 // 最近一次量到的 ADC 原始值
50 volatile unsigned int lastTempRaw;
51
52 int main(void)
53 {
54 WDTCTL = WDTPW | WDTHOLD; // 關掉 watchdog
55
56 initTemperatureSensor(); // 初始化溫度感測器 + ADC
57
58 while (1)
59 {
60 // 每次呼叫 readTemperature() 讀一次溫度 (raw)
61 lastTempRaw = readTemperature();
62
63 // 簡單 delay，避免讀太頻繁 (約 0.1 秒)
64 __delay_cycles(100000);
65 }
66 }
67
Problems 3
Output
Debug Output
Debug Console
Flash/FRAM usage is 384 bytes. RAM usage is 162 bytes.

實作結果(吹熱風後):



(2) Set up FRAM storage for the temperature data. Ensure that the temperature data collected is written into FRAM after every reading. Students should ensure that each time the system reads a new temperature value, it is stored into FRAM immediately. (10 pts)

變數:

```
#define LOG_SIZE 256   // 最多記 256 筆

// 在 FR4133 上，global 變數會放在 FRAM
unsigned int tempLog[LOG_SIZE];   // 每次量到的 ADC raw 值
unsigned int logIndex = 0;        // 指到下一筆要寫入的位置

volatile unsigned int lastTempRaw; // 最近一次讀到的 ADC 值
```

readTemperature():

```c
unsigned int readTemperature(void)
{
    ADCCTL0 |= ADCSC;                // 開始轉換
    while (ADCCTL1 & ADCBUSY) ;      // 等待轉換完成
    return ADCMEM0;                  // 回傳 raw 值
}
```

main():

```c
int main(void)
{
    WDTCTL = WDTPW | WDTHOLD;       // 關掉 watchdog

    initTemperatureSensor();        // 初始化 ADC + 溫度感測器

    while (1)
    {
        // 1. 讀一次溫度（raw ADC 值）
        lastTempRaw = readTemperature();

        // 2. 立刻把這筆資料寫進 FRAM 紀錄區
        tempLog[logIndex] = lastTempRaw;

        // 3. logIndex 前進一格，超過就從 0 重新開始（循環 buffer）
        logIndex++;
        if (logIndex >= LOG_SIZE)
            logIndex = 0;

        // 4. 簡單 delay，避免讀太頻繁（約 0.1 秒）
        __delay_cycles(100000);
    }
}
```

透過 **CCS Debug** 的 **WATCH** 觀察 **tempLog[]** 的內容會隨時間更新，證明每次量測後都有立刻寫入 **FRAM:**

DEBUG

emptyPr...

THREADS

CALL STACK

VARIABLES

WATCH

lastTempRaw: 689	0x002200
logIndex: 1	0x002202
tempLog[0]: 689	0x002000
tempLog[1]: 0	0x002002
tempLog[2]: 0	0x002004

main.c ✕ File not found

emptyProject > main.c > main

```c
#include <msp430.h>

#define LOG_SIZE 256   // 最多記 256 筆

// 在 FR4133 上，global 變數會放在 FRAM
unsigned int tempLog[LOG_SIZE];   // 每次量到的 ADC raw 值
unsigned int logIndex = 0;        // 指到下一筆要寫入的位置

volatile unsigned int lastTempRaw; // 最近一次讀到的 ADC 值


void initTemperatureSensor(void)
{
    // 先關掉 ADC，才可以改設定
    ADCCTL0 &= ~ADCENC;

    // 解鎖 PMM
    PMMCTL0_H = PMMPW_H;
    // 開啟 internal reference + temperature sensor
    PMMCTL2 |= INTREFEN | TSENSOREN;
    // 用預設參考電壓即可（對應 ADCSREF_1）
    PMMCTL0_H = 0;             // 鎖回 PMM

    // 設定 ADC
    ADCCTL0 = ADCSHT_5 | ADCON;    // 足夠長的 sample time + 開啟 ADC
    ADCCTL1 = ADCSHP;              // 使用 sampling timer
#ifdef ADCRES                        // 如果有 ADCRES，就設最高解析度
    ADCCTL2 = ADCRES;
#endif

    // 使用 A12（內部溫度感測器），參考電壓用 internal reference
    ADCMCTL0 = ADCINCH_12 | ADCSREF_1;

```

Problems 3 | Output | Debug Output ✕ | Debug Console

```
Flash/FRAM usage is 796 bytes. RAM usage is 676 bytes.
```

DEBUG

emptyPr...

THREADS

CALL STACK

VARIABLES

WATCH

lastTempRaw: 707	0x002200
logIndex: 2	0x002202
tempLog[0]: 689	0x002000
tempLog[1]: 707	0x002002
tempLog[2]: 0	0x002004

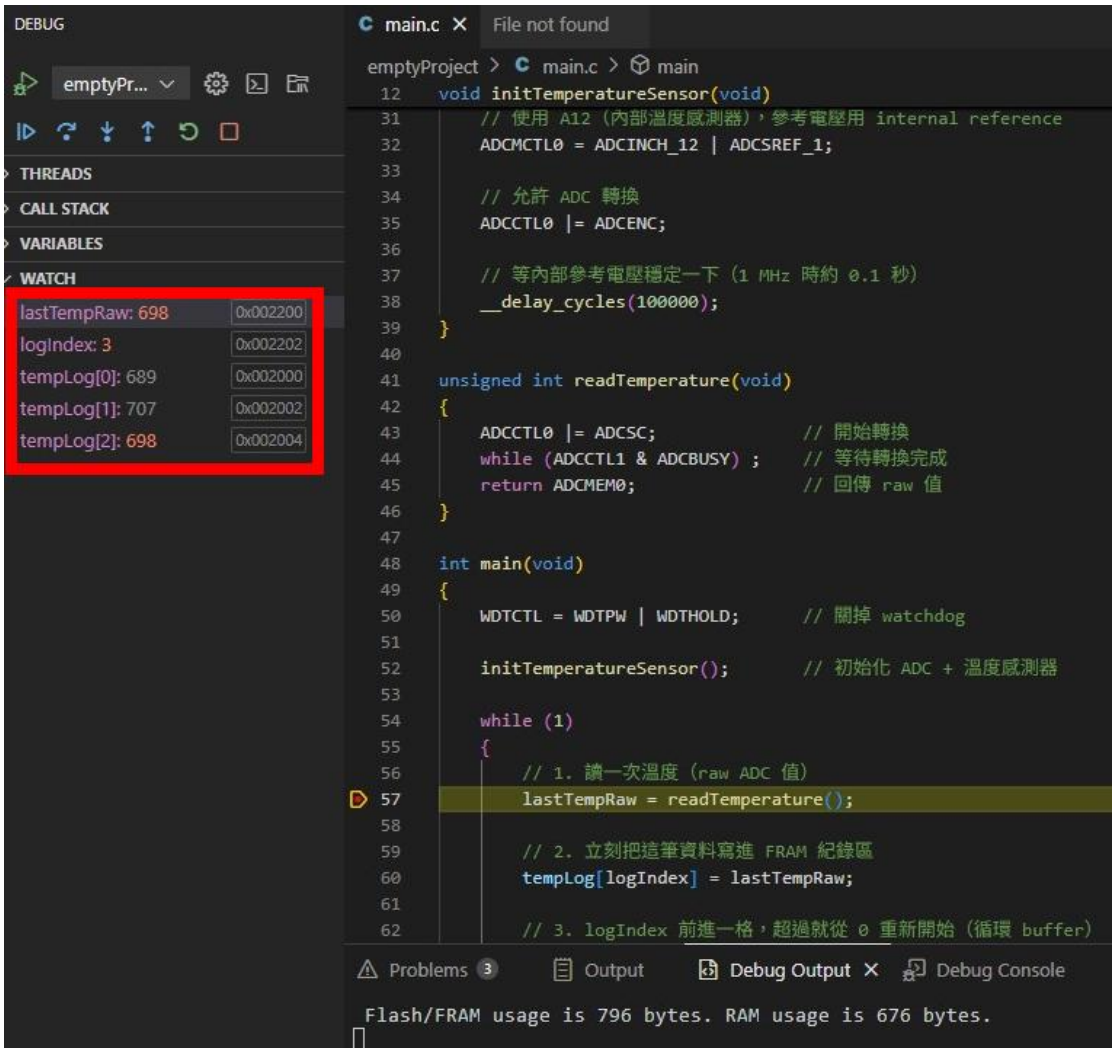
main.c ✕ File not found

emptyProject > main.c > main

```c
12  void initTemperatureSensor(void)
31      // 使用 A12（內部溫度感測器），參考電壓用 internal reference
32      ADCMCTL0 = ADCINCH_12 | ADCSREF_1;
33
34      // 允許 ADC 轉換
35      ADCCTL0 |= ADCENC;
36
37      // 等內部參考電壓穩定一下（1 MHz 時約 0.1 秒）
38      __delay_cycles(100000);
39  }
40
41  unsigned int readTemperature(void)
42  {
43      ADCCTL0 |= ADCSC;               // 開始轉換
44      while (ADCCTL1 & ADCBUSY) ;     // 等待轉換完成
45      return ADCMEM0;                 // 回傳 raw 值
46  }
47
48  int main(void)
49  {
50      WDTCTL = WDTPW | WDTHOLD;       // 關掉 watchdog
51
52      initTemperatureSensor();        // 初始化 ADC + 溫度感測器
53
54      while (1)
55      {
56          // 1. 讀一次溫度（raw ADC 值）
57          lastTempRaw = readTemperature();
58
59          // 2. 立刻把這筆資料寫進 FRAM 紀錄區
60          tempLog[logIndex] = lastTempRaw;
61
62          // 3. logIndex 前進一格，超過就從 0 重新開始（循環 buffer）
```

Problems 3 | Output | Debug Output ✕ | Debug Console

```
Flash/FRAM usage is 796 bytes. RAM usage is 676 bytes.
```



(3) Modify the system so that after a power loss, the last temperature data stored in FRAM is correctly retrieved and used for further processing. This ensures that no data is lost. Implement a power recovery mechanism where, upon reboot, the system checks FRAM for the last valid temperature value and uses it for processing. (10 pts)

writeLastTemp():

```
void writeLastTemp(unsigned int v)
{
    SYSCFG0 &= ~DFWP;    // unlock Data FRAM
    *lastTempPtr = v;
    SYSCFG0 |= DFWP;     // lock Data FRAM
}
```

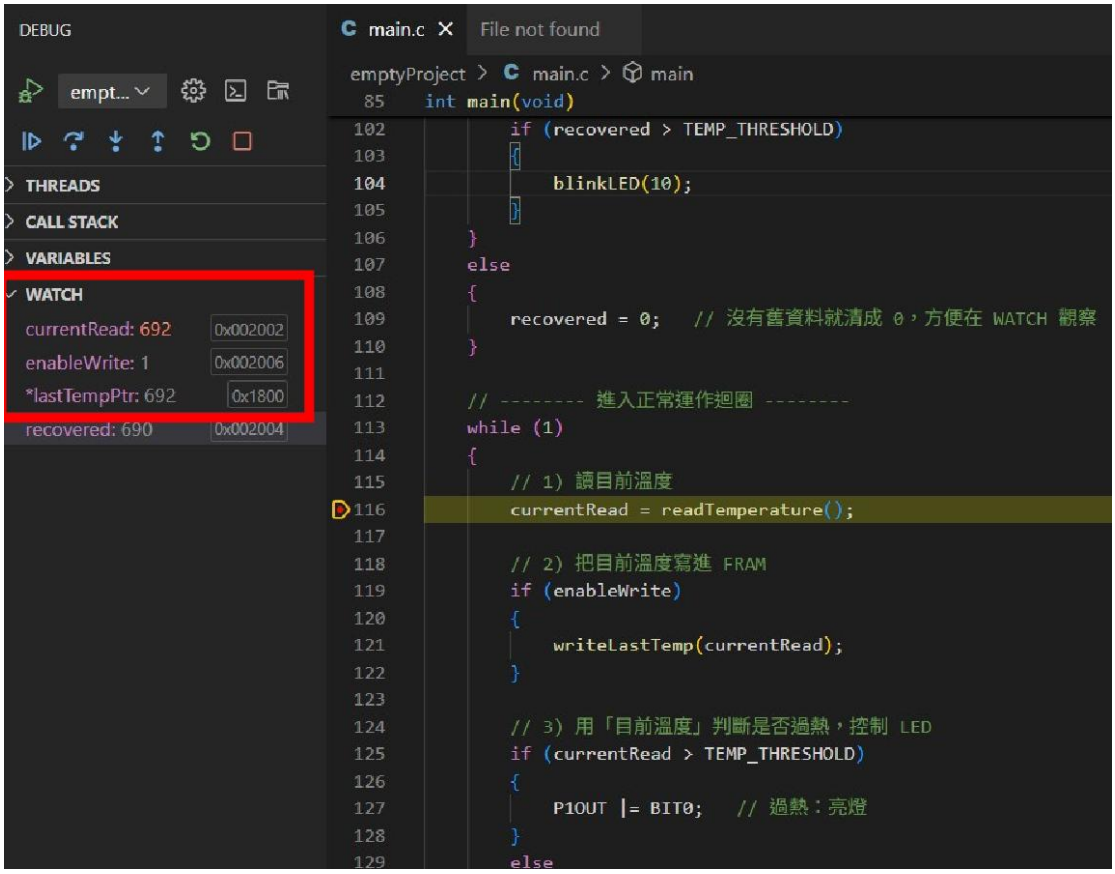
readLastTemp():

```c
unsigned int readLastTemp(void)
{
    return *lastTempPtr;
}
```

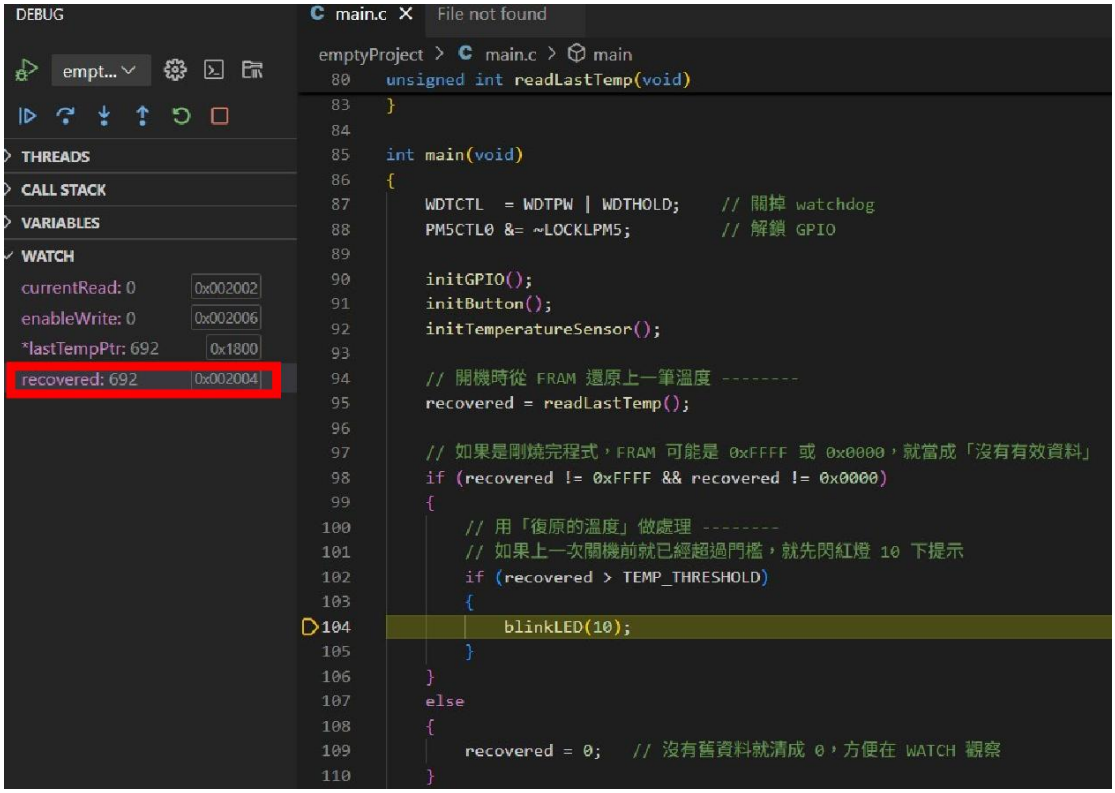
main()當中並且在 while 迴圈前的:

```c
// 開機時從 FRAM 還原上一筆溫度
recovered = readLastTemp();
```

斷電前測出溫度數值 692 並存進 FRAM 的*lastTempPtr:



斷電後 recovered 抓到斷電前的溫度:



(4)Use the temperature data retrieved from FRAM to implement a simple application, such as comparing the current temperature to a set threshold. If the temperature exceeds a specific threshold, trigger an LED on the board to indicate overheating. (10 pts)

門檻值:

```
#define TEMP_THRESHOLD 685
```

開機時，從 **FRAM** 讀回上一筆溫度並使用:

```
// 開機時從 FRAM 還原上一筆溫度 --------
recovered = readLastTemp();

// 如果是剛燒完程式，FRAM 可能是 0xFFFF 或 0x0000，就當成「沒有有效資料」
if (recovered != 0xFFFF && recovered != 0x0000)
{
    // 用「復原的溫度」做處理 --------
    // 如果上一次關機前就已經超過門檻，就先閃紅燈 10 下提示
    if (recovered > TEMP_THRESHOLD)
    {
        blinkLED(10);
    }
}
else
{
    recovered = 0;   // 沒有舊資料就清成 0，方便在 WATCH 觀察
}
```

Main()當中用現在量到的溫度控制 **LED**（過熱亮燈）**:**

```
if (currentRead > TEMP_THRESHOLD)
{
    P1OUT |= BIT0;   // 過熱：亮燈
}
else
{
    P1OUT &= ~BIT0;  // 未過熱：關燈
}
```

利用當前溫度對比門檻，過熱時真的會把 **LED** 打開

(此時 **currentRead=690>685):**

DEBUG

empt...

THREADS

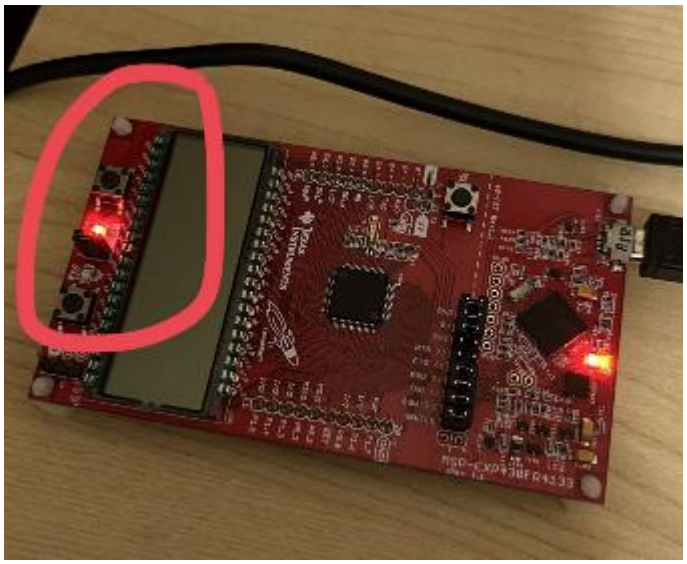
CALL STACK

VARIABLES

WATCH

currentRead: 690	0x002002
enableWrite: 0	0x002006
*lastTempPtr: 692	0x1800
recovered: 692	0x002004

main.c ✕ File not found

emptyProject > main.c > main

```c
int main(void)
{
    WDTCTL  = WDTPW | WDTHOLD;     // 關掉 watchdog
    PM5CTL0 &= ~LOCKLPM5;          // 解鎖 GPIO

    initGPIO();
    initButton();
    initTemperatureSensor();

    // 開機時從 FRAM 還原上一筆溫度 --------
    recovered = readLastTemp();

    // 如果是剛燒完程式，FRAM 可能是 0xFFFF 或 0x0000，就當成「沒有有效資料」
    if (recovered != 0xFFFF && recovered != 0x0000)
    {
        // 用「復原的溫度」做處理 --------
        // 如果上一次關機前就已經超過門檻，就先閃紅燈 10 下提示
        if (recovered > TEMP_THRESHOLD)
        {
            blinkLED(10);
        }
    }
    else
    {
        recovered = 0;   // 沒有舊資料就清成 0，方便在 WATCH 觀察
    }

    // -------- 進入正常運作迴圈 --------
    while (1)
```